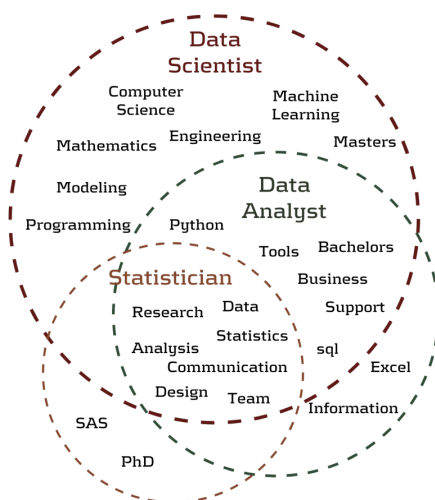


In recent job postings on LinkedIn and various platforms in the US, the term "data scientist" has become increasingly prevalent. Reflecting on my high school years, I had never encountered such titles, and now, there's a multitude of definitions and nuances surrounding what a data scientist truly is and the scope of their responsibilities. Before delving into my preferred definition, let me share my own understanding of the role: a "data scientist" is essentially a scientist of data, closely intertwined with the intricacies of data handling. For instance, data scientists excel in data correction, ensuring the rectification of data is conducted randomly without biases and aligns with the specific requirements of clients. If a client seeks information on the weight and height of individuals in the US, there is no need to collect data on their IQ. Data scientists tailor their approach to meet the unique needs of each client. Furthermore, they are tasked with data analysis, requiring a deep understanding of effective and efficient methods for extracting meaningful insights. As data analysis sometimes surpasses human capabilities, proficiency in programming is often a requisite skill. Additionally, data scientists are proficient in communicating their findings to clients, necessitating strong verbal and written communication skills. However, these thoughts remain somewhat insufficient as I acknowledge my limited knowledge of the field, not being a data scientist myself. In my exploration, I came across an article titled "A Data Science Approach to Defining a Data Scientist." The author utilized Natural Language Processing (NLP) and Machine Learning (ML) methods to analyze and correct terms extracted from web-scraped job postings related to "data scientist." The results of their analysis are presented below.



This definition of a data scientist is my favorite because it effectively highlights the distinctions from other well-known job titles, such as statisticians and data analysts. The accompanying diagram illustrates the diverse skill set that data scientists encompass, including machine learning, programming, and communication, among others. Data scientists are not confined to merely constructing

models using ML or other methods; they engage in client communication, comprehend the unique business models of each client, and propose optimal strategies for business growth. The emphasis on a Master's degree in data science aligns with the observed trend of data scientists typically possessing higher educational qualifications. I also explore two additional definitions. Microsoft's succinct definition describes a data scientist as someone who "collects, analyzes, and interprets big data to uncover patterns and insights, make predictions, and create actionable plans." While this definition is concise, it falls short by not explicitly acknowledging the crucial role of communication with clients. A more comprehensive definition should encompass communication skills and specific data science topics like AI, ML, and NLP, distinguishing data scientists from data analysts. Another definition from Oxford Languages by Google describes a data scientist as a "person employed to analyze and interpret complex digital data, such as the usage statistics of a website, especially to assist a business in decision-making." While this definition addresses business decision-making and emphasizes analyzing "complex" data, it lacks specificity regarding the methods used for analyzing such complex data, potentially blurring the distinction between a data scientist and a data analyst. Furthermore, neither definition touches upon the educational requirements, a crucial aspect given the intricate skill set demanded by data science. In summary, the evolving role of a data scientist necessitates a multifaceted skill set, extending beyond mathematical thinking. Proficiency in communication, domain-specific knowledge, and advanced programming skills are indispensable for modern data scientists. These skills often require a higher level of education, emphasizing the complexity and specialization inherent in the field.

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